

Fixing Graphical Glitches (White Textures) in UE4 with WINE: An Analytical Report

Introduction

Running Unreal Engine 4 (UE4) through Wine, a compatibility layer for running Windows applications on other operating systems like Linux, can present unique challenges, especially concerning graphical fidelity. A common issue reported by users is textures appearing as white, disrupting the visual integrity of the engine's output. This report analyzes the available information to pinpoint the root causes behind this problem and detail comprehensive steps toward resolution. This analysis prioritizes recent sources and delves into low-level system interactions, offering a blend of fundamental and sophisticated approaches to achieving graphical stability.

Problem Definition: White Textures in UE4 on Wine

The symptom is clear: textures in UE4 projects running with Wine render as a solid, often bright, white color instead of their intended appearance. This diminishes the visual quality and can hamper development workflows. While the issue could be a singular problem source, research indicates a multifactored issue affecting UE4 when run in a WINE environment. Textures and assets not being uploaded correctly is a commonality with issues involving white textures ([Failed to import a texture. - Issues & Bug Reporting \(Fab\) - Epic Developer Community Forums](#)).

Foundational Solutions: Addressing Graphics Configuration and Drivers

1. Graphics Driver Updates: Initial troubleshooting in Unreal Engine commonly centers on updating graphics drivers ([How to Get Rid of White Artifacts Icons in Unreal](#)) and ensuring that the project isn't exceeding the power of the current Graphics Card ([How to Get Rid of White Artifacts Icons in Unreal](#)). Given the environment is WINE, ensure mesa, a 3D Graphics library, is up-to-date along with the GPU Drivers ([Issues with OpenGL under Wine](#)). If the GPU is an Integrated Graphics card, it might be insufficient for the needs of the Engine ([Issues with OpenGL under Wine](#)).

2. Adjusting Graphics Settings: Overly ambitious graphics settings can lead to rendering issues, including artifacts and textures not loading correctly ([How to Get Rid of White Artifacts Icons in Unreal](#)). Within UE4, try the following adjustments:

- Resolution: Reduce the project's rendering resolution.
- Texture Quality: Lower the texture quality settings.
- Anti-Aliasing: Experiment with different anti-aliasing methods or disable it altogether. Disabling Temporal Anti-Aliasing (TAA) may help ([How to Get Rid of White Artifacts Icons in Unreal](#)).

Intermediate Investigation: Addressing Import Issues, Rendering Pipelines, and Asset Integrity

When fundamental solutions fail, it's essential to investigate deeper into the UE4 project and its interaction with Wine. Here are avenues to consider:

1. Texture Import Settings and Formats:

- Texture Format: The format of the original texture files can impact how they are rendered. The suggested best format from one user is to keep files as “.psd” while working on them, then save as “.tga” for UE4, and for UE4 also stick roughness, metallic, specular, and opacity if applicable into the RGBA of a .tga ([Which are the best texture file formats to import to UE4? - World Creation - Epic Developer Community Forums](#)).
- Import Options: When importing assets into UE4, specific options can affect texture appearance. When importing FBX files, make sure ‘Import Materials’ and ‘Import Textures’ are checked to ensure that the program is even importing the textures ([Fbx imports not importing textures - Asset Creation - Epic Developer Community Forums](#)).

2. Addressing Potential Triangulation Errors:

- **Problem:** Texture distortion or glitches (wavy and pixelated base color) can appear in UE4 even if the textures look fine in Maya or Substance Painter ([unreal-engine-4-texturing-glitch-distortion-help](#)).
- **Solution:** Triangulation errors might be to blame if the model wasn't exported with baked triangles before texturing ([unreal-engine-4-texturing-glitch-distortion-help](#)). Re-export the model from

its source software (e.g., Maya, Blender, 3ds Max) to guarantee properly baked triangles.

3. Derived Data Cache and Corrupted Assets:

- Derived Data Cache (DDC): Clear the DDC by deleting the contents of the DerivedDataCache folder within your project's Saved folder ([Solved] unreal engine missing texture)(<https://www.dragonflydb.io/error-solutions/unreal-engine-missing-texture>). Corrupted data in the DDC can lead to a variety of issues, including texture problems.
- Asset Integrity: Verify whether the textures are correctly showing up on another machine ([Solved] unreal engine missing texture)(<https://www.dragonflydb.io/error-solutions/unreal-engine-missing-texture>). Validate the project files' consistence if a Version Control is used.

Advanced Solutions: Wine Configuration, Graphics API Rendering, and Memory Management

The most resilient solutions tackle low-level interactions between Wine and UE4. Recent innovations in Wine itself suggest the most promising directions.

1. Graphics API Selection:

- OpenGL vs. Vulkan: Wine and UE4 can use either OpenGL or Vulkan for graphics rendering. Determine which API is being used, and try switching to see if it resolves the issue. While general OpenGL performance is a possibility, Vulkan might be the better choice ([Issues with OpenGL under Wine](#)).
- DXVK Integration: Direct3D to Vulkan translation layers like DXVK can improve performance and compatibility for some games and applications running under Wine ([Issues with OpenGL under Wine](#)). Trying DXVK might resolve some underlying issues with Direct3D rendering when running in WINE.

2. Wine Patches and Custom Builds:

- Out-of-tree patches: TensorWorks in collaboration with Epic Games, has been deploying Windows-specific Unreal Engine cloud workloads in Linux containers using the Wine compatibility layer ([Migrating Unreal Engine cook workloads to Linux with Wine](#)). Because target

platforms only support running under a Windows host platform, Wine allows for cloud deployment scenarios that provide cost savings and benefit from cloud native tooling ([Migrating Unreal Engine cook workloads to Linux with Wine](#)). Resources for unreal-engine-licensees to get going with WINE under cloud workloads is located in this Github repository: <https://github.com/EpicGamesExt/WineResources>.

3. Memory Management:

- RADV and Mesa Fixes: A Mesa Radeon Vulkan driver "RADV" works around Bugs for Unreal Engine 4 & 5, by addressing visual glitches and other on-screen artifacts appearing in Unreal Engine 5 Demos when using Mesa Radeon Vulkan Driver ([RADV Works Around Bugs For Unreal Engine 4 & 5](#)). This behavior can also be set manually by using the RADV_DEBUG=zerovram environment variable ([RADV Works Around Bugs For Unreal Engine 4 & 5](#)).
- Wine Memory Allocation: Wine needs to correctly approach memory allocation, ensuring that various assumptions made by Unreal Engine code under Windows will hold true under Wine ([Migrating Unreal Engine cook workloads to Linux with Wine](#)).

Conclusion and Recommended Action Plan Given the heterogeneous nature of why UE4 textures render as white under WINE, an all-encompassing strategy must be adopted. In summary, the issues stem from texture import settings, triangulation problems, data corruption, driver problems, and underlying WINE issues. Start with fundamental solutions, and work towards advanced solutions.

Recommended Action Plan:

1. **Driver and system libraries validation:** First ensure that both mesa, the graphics library, and the GPU driver, are valid and are up to date in the system libraries. This will solve underlying driver rendering issues, and is the most common problem ([Graphics Stack Released with NVK and RADV Driver Improvements - 9to5Linux](#)).
2. **Texture import and triangulation validation:** Once basic driver problems are out of the way, ensure that settings relevant to importing data are enabled. The options 'Import Materials' and 'Import Textures'

must be enabled to allow textures to be imported ([Fbx imports not importing textures - Asset Creation - Epic Developer Community Forums](#)). If that wasn't the issue, make sure that the model correctly exported, and baked properly. Re-export to ensure it did ([unreal-engine-4-texturing-glitch-distortion-help](#)).

3. **Implement Advanced solutions for WINE:** Once basic import settings are out of the way, ensure you are leveraging the proper advanced solutions for WINE, first by setting the mesa RADV variable to zero, and applying the relevant patches, to allow for better WINE performance ([Migrating Unreal Engine cook workloads to Linux with Wine](#)).

Following this ordered strategy ensures that basic settings are configured and validated, and that code patches under WINE are installed, to truly fix the issue of white textures under UE4.

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